

Alexander Arens

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EDUCATION

Purdue University
Bachelor of Science, Electrical and Computer Engineering

West Lafayette, IN
December 2027

PROFESSIONAL EXPERIENCE

Fuse Medical
Engineering Intern

Dallas, TX
May 2024 – Present

- Assumed lead engineering responsibilities, applying DFM to replace a machined assembly with a novel linkage, cutting costs 30x.
- Engineered parametric models for legacy surgical tools, improving ergonomic design in direct response to surgeon feedback.
- Redesigned SolidWorks CAD templates to improve efficiency and standardization.
- Maintain remote work arrangements and supply prototype components.

PERSONAL PROJECTS

- Full portfolio link <https://flukeychip.github.io/Project-Portfolio/>
- **Autonomous Drone Defense (Purdue AI Hackathon):** Lead a team to create an autonomous targeting system that prioritizes human and animal safety, while classifying identifying and “neutralizing” airborne threats
- **Scholarship Recipient – All-In Summit:** Selected as a scholarship recipient for the All-In Summit in California from a pool of thousands of applicants. Engaged with leaders in technology, medical, business, and government sectors. Invited to contribute to a medical AI startup.
- **6DOF Arm:** Drawing inspiration from the robotic arms available on campus, I have spent several weeks designing my own. The project is still in the CAD and design phase, but it is likely to my biggest project to date.
- **Large format laser engraver:** Designed and built a large-format (500 mm x 500 mm) laser engraver featuring automatic homing functionality and a two-factor safety system.
- **Rapidly deployable quadcopter:** Led a team to develop a drone with a compact, easily manufacturable design, a hot-swappable camera system, and ultrasonic sensor-based obstacle avoidance.
- **Smart Drink Dispenser Table:** Developed a device designed for discreet integration into a tabletop. Engineered the system to autonomously prepare beverages in response to user commands
- **Automated home security:** Lead a team of two other engineers to design a pan tilt camera system with IR flashlights, lasers and cameras to track and detect humans within a 360 radius. Extensively used cursor to program image recognition with closed loop visual feedback including PD control adaptive dead zones and platform specific optimizations
- **Monodirectional speaker:** Designed a circuit to drive ultrasonic speakers at 40 kHz and modulate them with an audible carrier wave. Produces directional audio output, audible only when the speaker is pointed directly at the listener.
- **Active orthosis (ASME):** Collaborated with the electronics division of an active orthopedic device that allowed patients with nerve or muscular damage to arms to restore movements using a multitude of sensors and actuators

RELEVANT SKILLS

- **Mechanical & CAD:** Proficient in Fusion360, SolidWorks, NX, KiCAD, and Autodesk Eagle.
- **Electronics:** PCB design, microcontrollers (ESP32, ATmega, RP2040), analog/digital analysis
- **Software & Programing:** FPGA (Verilog and ICE), MATLAB, Microsoft Excel, C, C++, Python
- **Prototyping:** Additive manufacturing (FDM, SLA, SLS), CNC machining, laser cutting, MIG welding

LEADERSHIP AND INVOLVEMENT

Eagle Scout | Phi Delta Theta project manager, Assistant House Manager | 3D Printing Club | PSP active controls | Purdue ASME

